Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++

Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Feature Engineering, Supervised & Unsupervised Learning, Model Evaluation, Statistical Analysis, Time-Series Analysis, Anomaly Detection

Deep Learning: PyTorch, TensorFlow/Keras, LSTM, Neural Networks, CNN

Data Analytics & Visualization: Exploratory Data Analysis, Power BI, Matplotlib, Plotly, Data Analysis

Backend & Deployment: FastAPI, Flask, Streamlit, REST APIs, Docker, Git/GitHub

Databases: SQL, SQLite, MySQL, MongoDB, Redis

Engineering Practices: Data Preprocessing, Model Validation, Reproducible Experimentation, PyTest, Test-Driven Development

EXPERIENCE

AI for Sustainability Virtual Intern* | LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered a Python analytics pipeline processing 50K+ transactions, combining statistical analysis, duplicate detection, vendor-velocity checks, HHI analysis, and Isolation Forest for automated anomaly detection.
- Designed a SQLite star-schema warehouse and composite 0–100 risk scoring pipeline, identifying 357 high/critical-risk transactions with automated analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures and automated pipeline tests to validate data quality, anomaly detection, and reporting reliability.

***Real-Time Stock Market Anomaly Detector* | LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

***CV-FL: Privacy-Preserving Federated Learning* | LINK**

- Developed a privacy-preserving federated learning framework combining model compression with lightweight verifiable aggregation for resource-constrained edge IoT environments.
- Achieved ~19.7× model compression while maintaining reliable aggregation, and implemented Merkle-tree commitments to detect tampered model updates.
- Evaluated robustness against malicious clients, achieving 100% tamper detection at ≥5% corruption and maintaining reliable operation with up to 30% malicious participants.

***AI-Powered Expense Receipt Scanner & Analytics Platform* | LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | LINK — 2026